

LAPORAN RESMI
“PENGENALAN LINGKUNGAN JAVA”



Disusun oleh:

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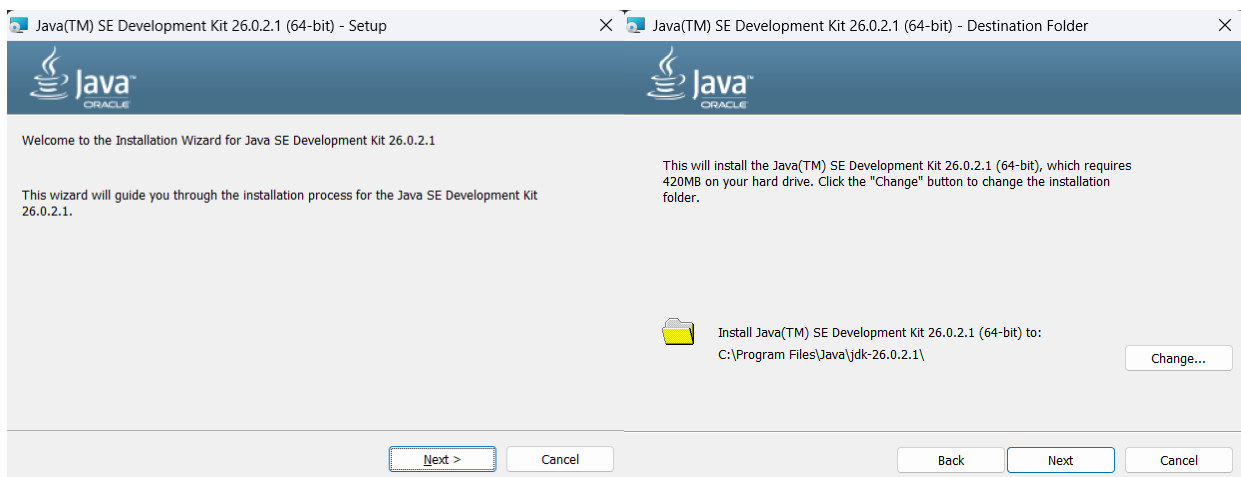
Tugas Pendahuluan

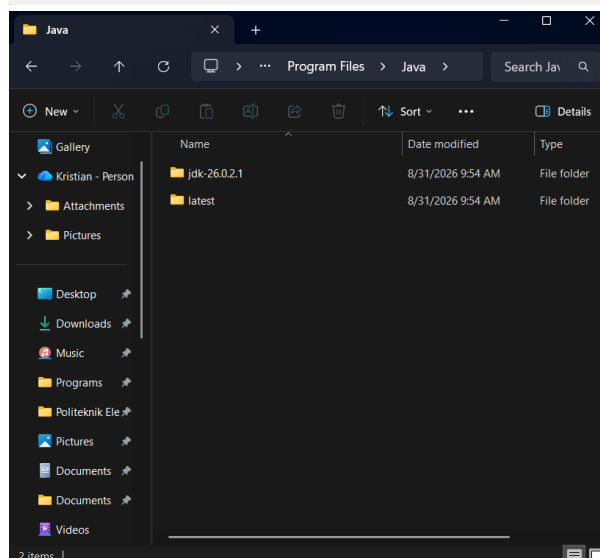
<https://docs.google.com/document/d/16z-JCAc8OW8eU4zQzQMOymN5w5LyZxMu/edit?usp=sharing&oid=100820287364085692120&rtpof=true&sd=true>

Percobaan

Percobaan 1: Instalasi Jawa

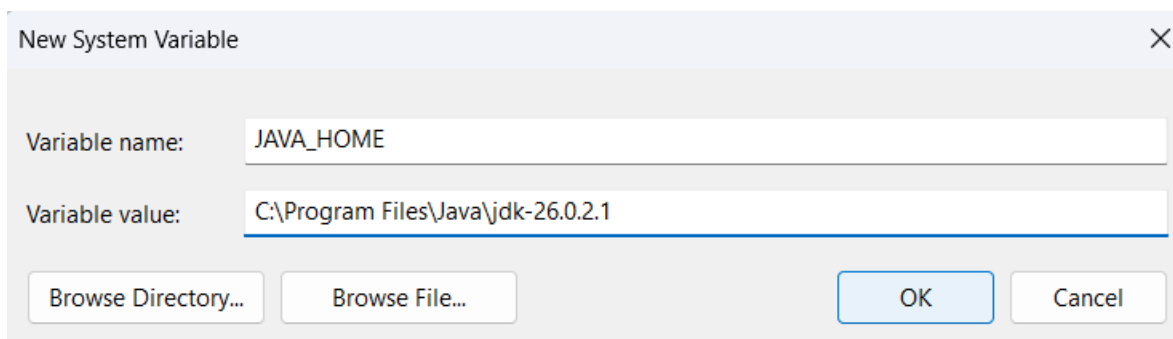
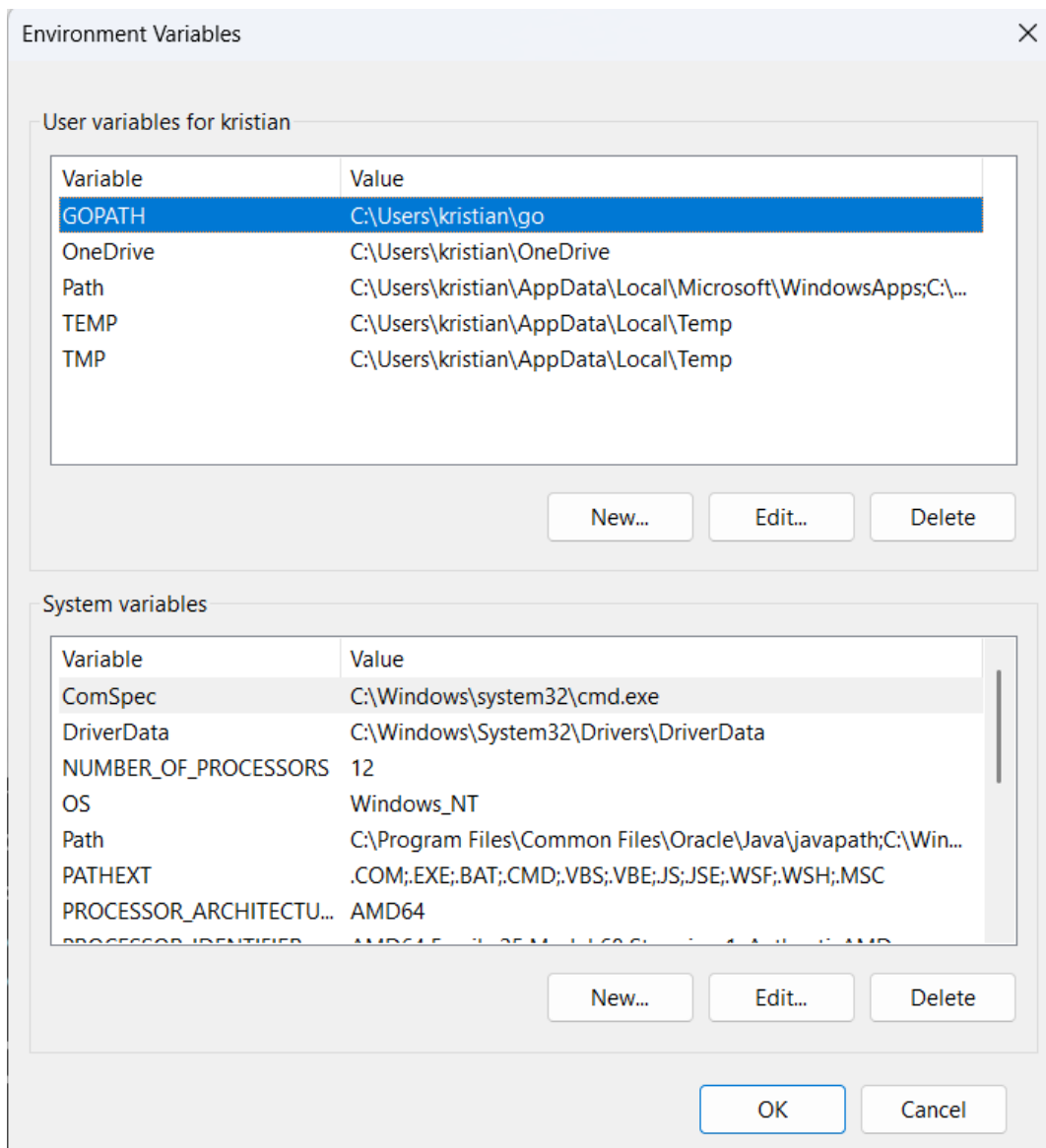
1. Instalasi JDK

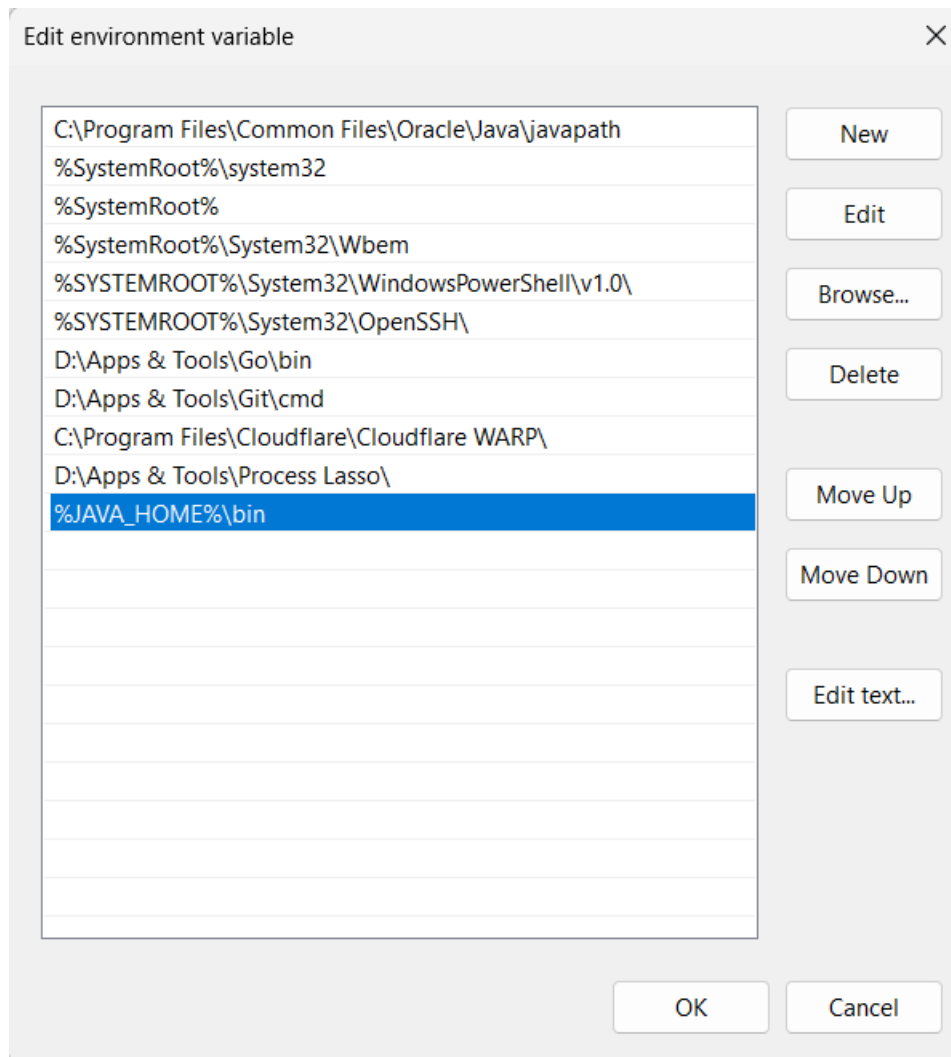




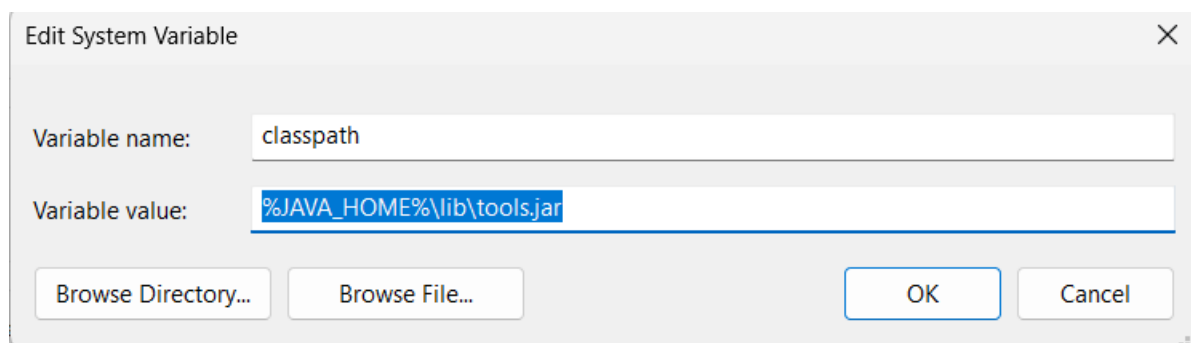
2. Pengesetan PATH

Control Panel → System and Security → System → Advance System Settings →
Enviroment Variable





3. Pengesetan CLASSPATH



4. Pengecekan hasil installasi

```

PS C:\Users\kristian> javac
Usage: javac <options> <source files>
where possible options include:
  @<filename>           Read options and filenames from file
  -Akey[=value]         Options to pass to annotation processors
  --add-modules <module>(,<module>)*
                        Root modules to resolve in addition to the initial modules,
                        or all modules on the module path if <module> is ALL-MODULE-PATH.
  --boot-class-path <path>, -bootclasspath <path>
                        Override location of bootstrap class files
  --class-path <path>, -classpath <path>, -cp <path>
                        Specify where to find user class files and annotation processors
  -d <directory>        Specify where to place generated class files
  -deprecation
                        Output source locations where deprecated APIs are used
  --enable-preview
                        Enable preview language features.
                        Also disables the 'preview' lint category.
                        To be used in conjunction with either -source or --release.
  -encoding <encoding>  Specify character encoding used by source files
  -endorseddirs <dirs>  Override location of endorsed standards path
  -extdirs <dirs>       Override location of installed extensions
  -g
                        Generate all debugging info
  -g:{lines,vars,source}
                        Generate only some debugging info
  -g:none
                        Generate no debugging info
  -h <directory>
                        Specify where to place generated native header files
  --help, -help, -?     Print this help message
  --help-extra, -X      Print help on extra options
  -implicit:{none,class}
                        Specify whether to generate class files for implicitly referenced files
  -J<flag>              Pass <flag> directly to the runtime system
  --limit-modules <module>(,<module>)*
                        Limit the universe of observable modules
  --module <module>(,<module>)*, -m <module>(,<module>)*
                        Compile only the specified module(s), check timestamps
  --module-path <path>, -p <path>
                        Specify where to find application modules
  --module-source-path <module-source-path>
                        Specify where to find input source files for multiple modules
  --module-version <version>
                        Specify version of modules that are being compiled
  -nowarn
                        Generate only mandatory warnings
  -parameters
                        Generate metadata for reflection on method parameters
  -proc:{none,only,full}
                        Control whether annotation processing and/or compilation is done.
  -processor <class1>[,<class2>,<class3>...]
                        Names of the annotation processors to run;

```

```

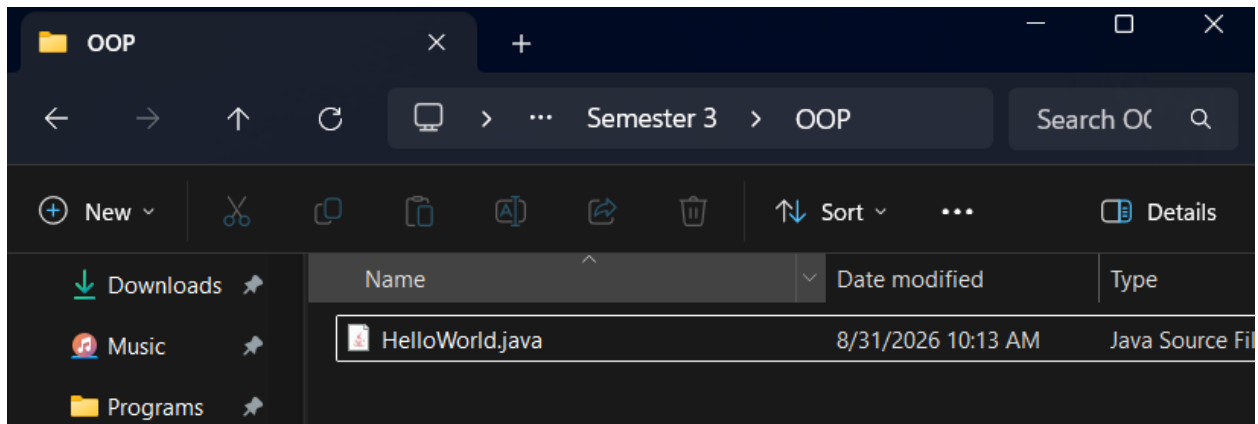
    bypasses default discovery process
--processor-module-path <path>
    Specify a module path where to find annotation processors
--processor-path <path>, -processorpath <path>
    Specify where to find annotation processors
-profile <profile>
    Check that API used is available in the specified profile.
    This option is deprecated and may be removed in a future release.
--release <release>
    Compile for the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
-s <directory>
    Specify where to place generated source files
--source <release>, -source <release>
    Provide source compatibility with the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
--source-path <path>, -sourcepath <path>
    Specify where to find input source files
--system <jdk>|none
    Override location of system modules
--target <release>, -target <release>
    Generate class files suitable for the specified Java SE release.
    Supported releases:
        8, 9, 10, ..., 24, 25, 26
--upgrade-module-path <path>
    Override location of upgradeable modules
-verbose
    Output messages about what the compiler is doing
--version, -version
    Version information
-Werror
    Terminate compilation if any warnings occur
-Werror:<key>(<key>)*
    Specify lint categories for which warnings should terminate compilation,
    separated by comma.
    Precede a key by '-' to exclude the specified category.
    Use --help-lint to see the supported keys.

```

PS C:\Users\kristian>

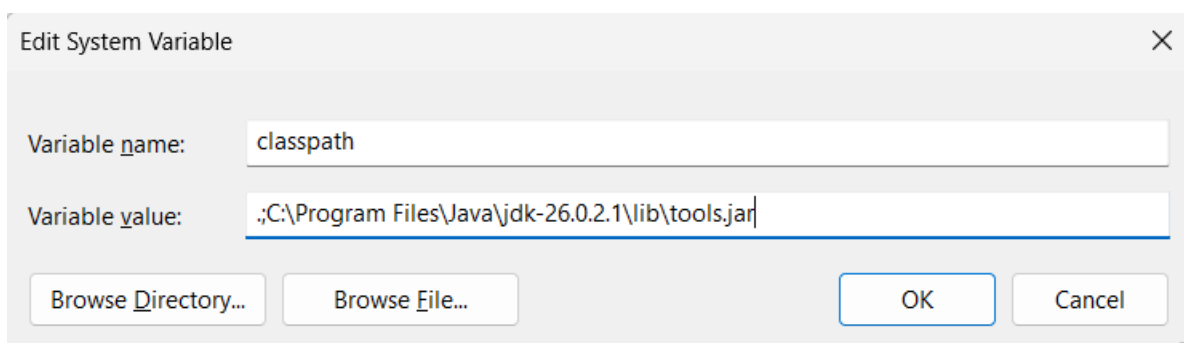
Percobaan 2 : Membuat Aplikasi Hello world

```
public class HelloWorld {  
    public static void main(String args[]) {  
        System.out.println("Hallo World...!");  
    }  
}
```



```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>java HelloWorld  
Error: Could not find or load main class HelloWorld  
Caused by: java.lang.ClassNotFoundException: HelloWorld  
  
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>
```

Problemnya ada di classpath, perbaiki classpath agar dia tidak mencari di satu folder saja saja tapi juga yang lainnya.



```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>java HelloWorld  
Hallo World...!
```

Sukses, program berjalan dengan lancar.

Percobaan 3 : Menganalisa dan Memperbaiki kesalahan

```
public class LuasKubus{  
    public static void main(String[] args){  
        int sisi = 5;  
        int luas = sisi*sisi*sisi;  
        System.out.println("Luas Kubus = " + luas);  
    }  
}
```

```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>javac LuasKubus.java
```

```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>java LuasKubus  
Luas Kubus = 125
```

Latihan

Latihan 1 : Menganalisa dan membenahi kesalahan program

```
public class LuasKubus{  
    public void main(String[] args){  
        double r = 5;  
        double luas = Math.PI * r * r;  
        double keliling = 2 * Math.PI * r;  
        System.out.println("Luas = " + luas);  
        System.out.println("Keliling = " + keliling);  
    }  
}
```

- Compile dan jalankan Program diatas.
- Catatlah error yang terjadi.
- Analisa dan perbaiki error tersebut.

```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>javac LuasKubus2.java  
LuasKubus2.java:1: error: class LuasKubus is public, should be declared in a file named LuasKubus.java  
public class LuasKubus{  
    ^  
1 error
```

Analisis:

1. error terjadi karena ada ketidaksesuaian nama file dan class, karena classnya public maka nama file harus sesuai.
2. beberapa sumber mengatakan bahwa bagian “public void main” bukanlah bagian yang lengkap, lebih baik diubah menjadi “public static void main” karena Java memanggil method main tanpa membuat objek dulu.
3. lebih baik nama class disesuaikan dengan rumusnya karena rumus tersebut merupakan rumus lingkaran bukan kubus.

Solusi:

1. nama file dan class disesuaikan
2. tambahkan “static”
3. nama class disesuaikan

```
public class LuasLingkaran{
    public static void main(String[] args){
        double r = 5;
        double luas = Math.PI * r * r;
        double keliling = 2 * Math.PI * r;
        System.out.println("Luas = " + luas);
        System.out.println("Keliling = " + keliling);
    }
}
```

E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>javac LuasLingkaran.java

E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>java LuasLingkaran

Luas = 78.53981633974483

Keliling = 31.41592653589793

Tugas

1. Buatlah program untuk konversi suhu Celcius menjadi Fahrenheit!

```
public class Convert{  
    public static void main(String[] args){  
        int c = 25;  
        double f = (c * (9.0 / 5.0)) + 32;  
        System.out.println("Celcius = " + c);  
        System.out.println("Fahrenheit = " + f);  
    }  
}
```

```
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>javac Convert.java  
  
E:\Politeknik Elektronika Negeri Surabaya (PENS)\Semester 3\OOP>java Convert  
Celcius = 25  
Fahrenheit = 77.0
```